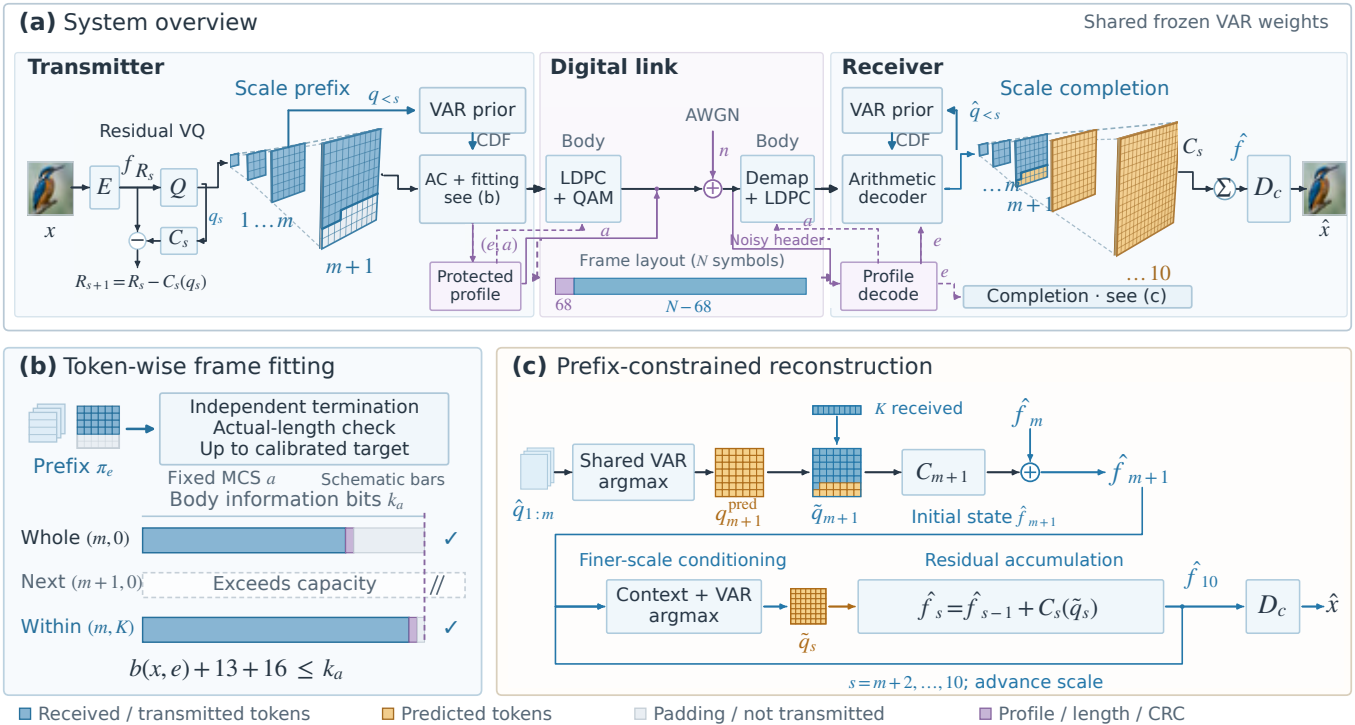

(a) System overview
Shared frozen VAR weights
Transmitter
Scale prefix
Residual VQ
VAR prior
CDF
AC + fitting see (b)
Protected profile
Digital link
AWGN
Body
LDPC + QAM
Body
Demap + LDPC
Noisy header
Frame layout (N symbols)
68
N − 68
Receiver
VAR prior
CDF
Arithmetic decoder
Profile decode
Scale completion
… 10
Completion · see (c)
(b) Token-wise frame fitting
Prefix π_e
Independent termination
Actual-length check
Up to calibrated target
Fixed MCS a
Schematic bars
Body information bits k_a
Whole (m, 0)
Next (m+1, 0)
Exceeds capacity
Within (m, K)
b(x, e) + 13 + 16 ≤ k_a
(c) Prefix-constrained reconstruction
K received
Shared VAR argmax
Initial state
Finer-scale conditioning
Residual accumulation
Context + VAR argmax
s = m+2, …, 10; advance scale
Received / transmitted tokens
Predicted tokens
Padding / not transmitted
Profile / length / CRC